ReLife Analytics

SALES, RETURNS & SUSTAINABILITY DASHBOARD

PROJECT DETAIL	DESCRIPTION
Course / Subject	Data Analytics and Visualization (DAV)
Project title	ReLife – Sales, Returns and Sustainability Analytics Dashboard
Primary platform	Microsoft Power BI
Data preparation	Cleaned project datasets; Power Query / data preparation workflow
Analytical focus	Sales performance, customer orders, product returns and environmental indicators

PROJECT GROUP MEMBERS

ROLL NUMBER	TEAM MEMBER
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An educational data analytics project that transforms cleaned sales and sustainability-related data into clear, interactive business insights.

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Executive Summary

ReLife Analytics is a data visualization project designed to bring important sales, returns and sustainability indicators into a single, easy-to-understand Power BI report. The project uses two cleaned datasets supplied by the team and focuses on presenting key metrics through KPI cards, charts and interactive slicers.

The first dataset includes order-level and product-level fields such as Order_ID, Product_ID, User_ID, Order_Date, Product_Category, Product_Price, Order_Quantity, Discount_Applied, Shipping_Method, Payment_Method, User_Age, User_Gender, User_Location, Return_Status, Return_Reason, Days_to_Return, Order_Value, Return_Cost, Profit_Loss, CO2_Emissions, Packaging_Waste, CO2_Saved and Waste_Avoided.

The second cleaned dataset is also part of the Power BI model. Its exact column list and row count were not available in the materials reviewed for this report; therefore, no unverified field names or numerical findings are invented. The report describes the intended analysis and marks data-dependent results for completion after validation in Power BI.

The planned one-page dashboard summarizes revenue, transaction volume, purchased item quantity and environmental impact, then provides visual analysis of monthly revenue, category performance, return reasons, return costs and sustainability indicators. Interactive slicers are intended to help users explore the report by date, category, location and return status where the data model supports those filters.

Report status: dashboard design and construction are in progress. Final KPI values, chart findings and screenshots should be inserted after the report page is completed and checked against the underlying datasets.

Report organization

The report covers the problem statement, objectives, dataset description, preparation workflow, analytical methodology, dashboard design, implementation, limitations, future enhancements and references.

1. Problem Statement and Motivation

1.1 Business context

E-commerce operations generate data about orders, products, customers, discounts, shipping, payments and returns. When these details are reviewed only as rows in spreadsheets, it can be difficult for users to identify sales patterns, compare product categories, understand return reasons or evaluate environmental indicators.

1.2 Problem statement

The project needs a consolidated and user-friendly way to explore sales performance, product returns and sustainability-related measures from two cleaned datasets. The information should be understandable to both technical and non-technical users and should support quick comparisons without requiring them to inspect raw tables.

1.3 Proposed solution

ReLife Analytics uses Microsoft Power BI to create a single-page interactive dashboard. KPI cards provide a quick overview, charts explain patterns and comparisons, and slicers allow users to filter the report. The dashboard uses a consistent green-and-white visual theme to reinforce the sustainability focus while maintaining readability.

1.4 Project value

The dashboard is intended to support data-driven understanding of revenue trends, category contribution, return patterns, return-related costs and environmental measures. It is an analytical reporting tool; it does not automatically prove that a product or business practice is environmentally sustainable.

2. Project Objectives and Scope

The main objectives are to:

• Combine the two cleaned project datasets in a Power BI report model where appropriate.
• Summarize important sales and sustainability measures using KPI cards.
• Explore revenue changes over time and revenue contribution by product category.
• Examine return reasons and return-related costs using the available return fields.
• Compare CO■ emissions, CO■ saved, packaging waste and waste avoided with correct units and aggregation.
• Add interactive slicers and clear visual labels to make the report easy to use.
• Present the results in one organized report page.

2.1 Scope of the current implementation

IN SCOPE	NOT CLAIMED AS COMPLETED
Two cleaned datasets imported into Power BI; a one-page dashboard; KPI cards; sales, returns and sustainability visuals; slicers and formatting.	Predictive AI/ML, automated recommendations, live data feeds, causal conclusions or verified business improvements unless separately implemented and tested.

3. Dataset Description and Data Fields

3.1 Dataset 1 – Order, product, customer, returns and sustainability fields

The first cleaned dataset contains the following fields as provided by the team:

FIELD	ROLE / INTERPRETATION
Order_ID	Order identifier; useful for distinct transaction counts.
Product_ID	Product identifier.
User_ID	Customer identifier; handle as an ID, not a measure.
Order_Date	Date field for time-based analysis.
Product_Category	Category for product and revenue comparisons.
Product_Price	Numeric product price.
Order_Quantity	Quantity of items ordered.
Discount_Applied	Discount value or rate; confirm its unit before aggregation.
Shipping_Method	Shipping category.
Payment_Method	Payment category.
User_Age	Customer age in years, if recorded as such.
User_Gender	Customer demographic category.
User_Location	Customer location category.
Return_Status	Indicates return status.
Return_Reason	Reason category, where populated.
Days_to_Return	Number of days until return, where applicable.
Order_Value	Order value; verify whether gross or net of discounts.
Return_Cost	Return-related cost; confirm unit and grain.
Profit_Loss	Profit/loss measure; confirm calculation definition.
CO2_Emissions	CO■ emissions measure; preserve its source unit.
Packaging_Waste	Packaging waste measure; preserve its source unit.
CO2_Saved	CO■ saved measure; preserve its source unit.
Waste_Avoided	Waste avoided measure; preserve its source unit.

3.2 Dataset 2 – Additional cleaned dataset

The second cleaned dataset was imported into the Power BI project according to the team’s progress. Its exact filename, field names, row count and relationship keys should be copied from Power BI's Data view and Model view before final submission. This report intentionally avoids guessing those details.

3.3 Data model considerations

Before combining tables, inspect the grain of each dataset and identify a reliable shared key. If both tables contain repeated Order_ID values, a direct many-to-many relationship may produce duplicated totals. Prefer a suitable dimension table or a clearly defined one-to-many relationship. Keep measures in the table whose grain and meaning they represent.

4. Data Preparation and Analytical Methodology

4.1 Data preparation

The project uses cleaned datasets as inputs to Power BI. Before analysis, confirm that identifiers are stored as text, Order_Date is stored as a date, quantities and counts use whole-number types, and financial or environmental measures use numeric types. Text categories should be standardized so that spelling or whitespace differences do not create duplicate categories.

4.2 Recommended validation checks

• Check row counts and column names for both cleaned datasets.
• Check missing values in key fields such as Order_ID, Order_Date and Product_Category.
• Check duplicate records and confirm whether repeated order IDs are expected at product-line level.
• Verify that Order_Value, Return_Cost and Profit_Loss have consistent definitions and units.
• Verify the units and meaning of CO2_Emissions, CO2_Saved, Packaging_Waste and Waste_Avoided.
• Confirm relationships and cross-filter direction in Model view.

4.3 Analytical measures

The following are suggested measure definitions. Replace table names with the exact names shown in the Power BI Fields pane and validate each measure against the dataset grain.

MEASURE	SUGGESTED LOGIC / CHECK
Total Revenue	SUM of the verified revenue field (for example, Order_Value or the team's approved Final Revenue measure). Avoid summing both if they represent the same amount.
Total Transactions	Distinct count of Order_ID, not row count, if one order can contain multiple product rows.
Total Purchased Items	SUM of Order_Quantity, if each row represents a product line.
Return Rate	Returned distinct orders divided by eligible distinct orders. Filter using Return_Status and define the denominator.
Total CO■ Saved	SUM of CO2_Saved, only after confirming unit and record grain.
Total Waste Avoided	SUM of Waste_Avoided, only after confirming unit and record grain.

4.4 Example DAX patterns

Use these as patterns only; adjust the table and column names to match the actual model:

Total Transactions

```
Total Transactions = DISTINCTCOUNT('Sales'[Order_ID])
```

Total Purchased Items

```
Total Purchased Items = SUM('Sales'[Order_Quantity])
```

Total Revenue

```
Total Revenue = SUM('Sales'[Order_Value])
```

A Return Rate measure should not be created until the correct return flag/status values and denominator are confirmed. Avoid using a column name such as Purchased Item Count unless it exists exactly in the imported dataset.

5. Exploratory Analysis and Dashboard Design

5.1 Planned KPI cards

KPI CARD	PURPOSE / VALIDATION
Total Revenue	Summarizes the selected revenue measure. Confirm whether revenue is gross, discounted or net.
Total Transactions	Counts unique orders using Order_ID.
Total Purchased Items	Sums Order_Quantity if the field represents item quantity.
Return Rate (%)	Shows returned orders as a percentage of eligible orders; requires validated Return_Status logic.
Total CO■ Saved	Summarizes CO2_Saved using the correct unit.
Total Waste Avoided	Summarizes Waste_Avoided using the correct unit.

5.2 Planned charts

VISUAL	FIELDS / QUESTION ANSWERED
Line chart – Monthly Revenue Trend	Order_Date on the time axis; verified revenue measure on the value axis. Shows changes over time.
Clustered bar chart – Revenue by Product Category	Product_Category and verified revenue measure. Compares category contribution.
Donut chart – Product Return Reasons	Return_Reason as legend; count of returned Order_ID records/orders as values. Filter to returned records.
Column chart – Return Cost by Product Category	Product_Category and SUM(Return_Cost). Compares return-related cost by category.
Clustered column chart – Sustainability Impact	Compare CO2_Emissions and CO2_Saved only if units and scale are compatible; use separate visuals for packaging waste and waste avoided if units differ.

5.3 Interactive slicers

The proposed slicers are Order Date, Product Category, User Location and Return Status. A slicer should only be expected to filter a visual when the data model has a valid relationship or the visual uses fields from the same table. Test each slicer with every visual before submission.

5.4 Layout and visual design

The one-page layout places the ReLife Analytics title at the top, six KPI cards beneath it, sales and return charts in the central area, sustainability analysis below, and slicers in a clearly visible filter strip. The chosen palette uses dark green (#174B3B), medium green (#21865B), light green and white. Use consistent font sizes, readable data labels, concise titles and meaningful units.

5.5 User interaction

Users should be able to select a date range, category, location or return status and observe relevant visuals update. Tooltips can reveal additional context without overcrowding the report. Use Edit interactions to confirm whether selections cross-filter or highlight other charts as intended.

6. Implementation, Tools and Limitations

TOOL	ROLE IN THE PROJECT
Microsoft Power BI Desktop	Import datasets, manage relationships, create DAX measures, build KPI cards/charts/slicers and format the report.
Power Query	Inspect data types, rename fields, standardize categories and perform any additional query transformations required.
DAX	Define measures such as revenue, distinct transactions, quantities and validated ratios.
Cleaned CSV / spreadsheet datasets	Provide the prepared source data for analysis.

6.1 Current implementation status

Based on the project conversation, the two cleaned datasets have been imported into Power BI and the team has begun building a single-page dashboard. The title/header and KPI-card/chart construction have been discussed. The final completed report page and verified numerical results were not supplied with the reference material; therefore this report describes the intended design rather than claiming that every visual is finished.

6.2 Limitations and assumptions

• The exact schema and record count of the second dataset have not been documented in the information available for this report.
• Final KPI values and chart findings are not stated because the cleaned source files and completed dashboard output were not provided for verification.
• The meaning and unit of Discount_Applied, Return_Cost and environmental measures must be confirmed before interpreting totals.
• A returned-order count is only valid when Return_Status is present and its values are understood.
• Revenue totals can be overstated if order-level values are repeated across multiple product rows.
• A single combined chart can mislead if emissions and waste measures use different units or scales.

6.3 Responsible interpretation

The dashboard provides descriptive analytics. It can show patterns and associations in the supplied data, but it does not establish why a trend occurred or prove that an intervention caused a change. Environmental metrics should be reported with their units, definitions and data source.

7. Conclusion and Future Enhancements

ReLife Analytics aims to make sales, returns and sustainability data easier to understand through a consolidated Power BI dashboard. By combining KPI cards, time-based and category-based charts, return analysis and environmental indicators, the report is designed to help users explore key patterns in a clear and interactive way.

The project's value depends on sound data modeling and accurate measure definitions. Distinct order counts, revenue aggregation, return-rate denominators and environmental units must be validated before final values are interpreted. A well-designed dashboard should prioritize correctness and clarity over the number of visuals.

7.1 Future enhancements

- Add a documented data dictionary for both cleaned datasets.
- Validate all relationships and DAX measures against source totals.
- Add drill-through to order or category detail if needed.
- Add informative tooltips for return reasons and sustainability metrics.
- Include a date comparison or year-over-year analysis if the dataset spans enough time.
- Add data refresh documentation and a clear source/version record.
- Capture the final Power BI dashboard and add it to the report as an implementation figure.

7.2 Final conclusion

The ReLife project demonstrates how cleaned operational data can be transformed into an accessible business intelligence report. The planned dashboard brings together sales performance, product returns and sustainability measures in one view. With final data validation, completed visuals and documented findings, it can serve as a useful academic demonstration of data preparation, analysis and interactive visualization.

8. References and Data Sources

1. Cleaned Dataset 1 supplied by the ReLife project team; field list documented in this report.
2. Cleaned Dataset 2 supplied by the ReLife project team; exact filename and schema to be added by the team.
3. Microsoft Power BI documentation – data modeling, report visuals, slicers and DAX.
4. Microsoft Power Query documentation – data import and transformation.
5. Project notes and dashboard design decisions recorded by the ReLife team.

Appendix A – Dashboard Build Checklist

ITEM	CHECK BEFORE FINAL SUBMISSION
Data model	Confirm table names, keys, relationship cardinality and filter direction.
Revenue	Confirm correct revenue field and whether it is repeated at order or line level.
Transactions	Use distinct Order_ID where orders have multiple rows.
Returns	Confirm Return_Status categories and define the return-rate denominator.
Sustainability	Confirm units and aggregation for CO■ and waste fields.
Charts	Check axis titles, chronological month sorting, labels, tooltips and sort order.
Slicers	Test Date, Category, Location and Return Status across all relevant visuals.
Presentation	Check alignment, readable fonts, consistent colours, page fit and no overlapping visuals.
Evidence	Insert screenshots of the final dashboard and record verified KPI values/findings.

Appendix B – Project Group Members

ROLL NUMBER	TEAM MEMBER
2503A52605	S. Sri Chandana
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Prepared as an academic project report. Replace the noted validation items with confirmed values and screenshots before final submission.